THE VIRTUAL MEET EXPERIENCE

2017 - 2018
HS VIRTUAL CHALLENGE
STATE QUALIFIERS' MEET

COMPUTER SCIENCE

DO NOT OPEN TEST UNTIL TOLD TO DO SO

2017-2018 Virtual Challenge State Qualifiers' Meet – Computer Science

Note: Correct responses are based on Java, **J2sdk v 1.8.x**, from Sun Microsystems, Inc. All provided code segments are intended to be syntactically correct, unless otherwise stated (i. e. `error` is an answer choice) and any necessary Java 2 Standard Packages have been imported. Ignore any typographical errors and assume any undefined variables are defined as used. **For all output statements, assume that the System class has been statically imported…** ***import static java.lang.System.*;***

QUESTION 1

What is 11111_2 - 10111_2 ?

A. 01010_2 B. 5_{10} C. 10_8 D. 2_6 E. 110_2

QUESTION 2

What is output by the code to the right?

A. 26 B. 27

C. 28 D. 25

E. There is no output due to an error.

```
out.println(3 * 3 * 3 + 4-5);
```

QUESTION 3

What is output by the code to the right?

A.
```
*----*----*----
4.33|3.44|5.67
```

B.
```
*----*----*----
5.67|4.33|3.44
```

C.
```
*----*----*----
3.44|4.33|3.44
```

D.
```
*----*----*----
3.44|4.33|5.67
```

E. There is no output due to an error.

```
out.println("*----*----*----");
out.printf("%2$.2f|%.2f|%.2f"
                  ,4.3333,3.4445,5.666);
```

QUESTION 4

What is output by the code to the right?

A. `baniag`

B. `banian`

C. `Zog1ofalg`

D. `lbaniag`

E. There is no output due to a syntax error.

```
String h="Zog1ofAlbania";
h=h.substring(8);
h+=h.charAt(2);
out.println(h);
```

QUESTION 5

How many combinations of variables a, b, c will result in d being set to the value true?

A. 3 B. 5

C. 2 D. 6

```
//assume a,b, and c are
//variables of type boolean

boolean d=a & b | c;
```

QUESTION 6

What is output by the code to the right?

A. -2147483648 B. 2147483647

C. -2147483649 D. 2147483646

E. There is no output due to an error.

```
int k=Integer.MIN_VALUE;
int h=Math.decrementExact(k);
out.println(h);
```

QUESTION 7

Which of the lines to the right first causes a syntax error?

A. //1
B. //2
C. //3
D. //5
E. None of the above cause an error.

```
Double[]d=new Double[5];
d[0]=Integer.MAX_VALUE;  //1
d[1]=Double.INFINITY;  //2
d[2]=Float.BYTES;  //3
d[3]=Double.MAX_VALUE;  //4
d[4]=Float.MAX_VALUE;  //5
```

QUESTION 8

What is the output by the code to the right?

A. 124
B. 14
C. 24
D. 13
E. There is no output due to an error.

```
int y=3;
if(y==3 || ++y>3)
 out.print(1);
if(y==4)
 out.print(2);
out.print(y);
```

QUESTION 9

What is the output by the code to the right ?

A. 182
B. 210
C. 240
D. 105
E. There is no output due to a syntax error.

```
String s="";
for(int y=0;y<14;y++) {
 for(int x=0;x<=y;x++)
    s+="$$";
}
out.println(s.length());
```

QUESTION 10

What is the output by the code to the right ?

A. 1
B. 2
C. 3
D. 4
E. 5

```
int[] a={1,4,3,5,2,0};
int y=a[a[a[a[a[0]]]]];
out.print(a[a[a[a[a[y]]]]]);
```

QUESTION 11

Which of the following could read in the following data for calculation purposes?

2147483648

A. file.nextInt()
B. file.nextDouble()
C. file.nextLine()
D. More than one of the above.
E. None of the above.

QUESTION 12

What is the runtime of the code to the right in Big O notation ?

A. O(n*n)
B. O(logn)
C. O(n)
D. O(nlogn)
E. O(1)

```
int h=0,n=12;
for(int y=0;y<n*2;y++) {
 for(int k=n;k<n+10;k*=2)
    h++;
}
```

QUESTION 13

Which of the operations to the right has the lowest precedence?

A. III
B. IV
C. I
D. II
E. V

```
I. +(unary)
II. >>=
III. >>>
IV. |
V. ||
```

QUESTION 14

What is the output by the code to the right ?

A. `-2147483648`
B. `2147483648`
C. `-2147483647`
D. `2147483649`
E. No output due to an error.

```
System.out.println(Integer.MAX_VALUE+1);
```

QUESTION 15

What is the output by line marked `//1` in the code to the right ?

A. `2`
B. `0`
C. `5`
D. There is no output due to a syntax error.
E. There is no output due to a runtime error.

QUESTION 16

What is the output by the line marked `//2` in the code to the right ?

A. `[2, 4, 5, 8, 8, 3]`
B. `[2, 8, 8, 12, 3]`
C. `[2, 8, 12, 8, 3]`
D. `[2, 12, 8, 8, 3]`
E. There is no output due to an error.

```
ArrayList a=new ArrayList();
a.add(9);
a.add(8);
a.add(0,12);
a.add(0,2);
a.set(2, 5);
out.println(a.remove(2)); //1
a.add(8);
a.add(3);
Collections.sort(a.subList(1, 3));
out.println(a); //2
```

QUESTION 17

What is the output by the code to the right ?

A. `5061`
B. `5051`
C. `4961`
D. `4951`
E. There is no output due to an error.

```
int tot=0;
for(int y=1;y<=100;y++) {
 tot+=y;
}
for(int h=0;h<=10;h++)tot++;
out.println(tot);
```

QUESTION 18

Which of the following could **NOT** replace **`<1*>`** in the code to the right so that the output is in the format shown in the answers to question 19?

A. `out.printf("%tB %<td, %tY",g);`
B. `out.printf("%tB %<td, %tY",g,g);`
C. `out.printf("%tB %<td, %1$tY",g);`
D. `out.printf("%tB %<td, %tY",g,g,g);`
E. More than one of the above.

QUESTION 19

Assuming **`<1*>`** is correctly filled, what is output by the client code to the right?

A. `March 29, 2013`
B. `February 29, 2013`
C. `March 30, 2013`
D. `February 30, 2013`
E. There is no output due to an error.

```
GregorianCalendar g;
g=new GregorianCalendar();
g.set(g.DATE, 26);
g.set(g.MONTH,2);
g.set(g.YEAR,2013);
g.add(g.HOUR, 66);

<1*>
```

QUESTION 20

What is the big O runtime of deleting an item from a standard array?

A. O(n) B.O(n^2) C.O(log(n))

D.O(n log(n)) E. O(1)

QUESTION 21

What is the runtime of the algorithm that is implemented in the code to the right, assuming N is the length of parameter arr?

	Best	Average	Worst
A.	O(N^2)	O(N^2)	O(N^2)
B.	O(NlogN)	O(NlogN)	O(NlogN)
C.	O(NlogN)	O(NlogN)	O(N^2)
D.	O(N)	O(N^2)	O(N^2)
E.	O(NK)	O(NK)	O(NK)

```
void fun(int arr[]){
 int n = arr.length;
 for (int i = 0; i < n-1; i++)
      for (int j = 0; j < n-i-1; j++)
          if (arr[j] > arr[j+1])
          {
              int temp = arr[j];
              arr[j] = arr[j+1];
              arr[j+1] = temp;
          }
}
```

QUESTION 22

What is the output by the line marked `//q22` in the client code to the right?

A. `6`

B. `1`

C. `5`

D. `4`

E. This is no output due to an error.

QUESTION 23

What is the output by the line marked `//q23` in the client code to the right?

A. `11`

B. `12`

C. `1`

D. `10`

E. This is no output due to an error.

QUESTION 24

If the commented line in method mystery is uncommented, what is the output by the line marked `//q22` in the client code to the right?

A. `[1, 1, 1, 2, 3, 4, 5]5`

B. `[1, 1, 0, 1, 2, 2, 1]5`

C. `[1, 1, 1, 2, 3, 4, 5]1`

D. `[1, 1, 0, 1, 2, 2, 1]1`

E. This is no output due to an error.

```
public static int mystery(int k) {
  int[] c= {1,3,4,5};
  int[] d=new int[k+1];
  d[0]=1;
  for(int cc:c) {
    for(int y=0;y<d.length;y++) {
      if(cc<=y && d[y-cc]>0)
       d[y]+=d[y-cc];
    }
  }
  // out.print(Arrays.toString(d));
  return d[k];
}


/////client code//////////////////////
out.println(mystery(6));  //q22
out.println(mystery(10));  //q23
```

QUESTION 25

Which of the following does not match the regex patttern to the right?

A. `""`

B. `"w1ww1ww1ww1ww1wd2d2"`

C. `"e3d33333333"`

D. `"w1ww1wa7"`

E. This is no output due to an error.

```
(\\w\\d\\w)*([^aeiou]\\d)*
```

QUESTION 26

What is output by the code to the right?

A. `62` B. `82`

C. `94` D. `74`

E. There is no output due to an error.

```
BigInteger b=new BigInteger("78");
b=b.flipBit(4);
out.println(b);
```

QUESTION 27

Which of the following is a legal instantiation?

A. `String g@sb=new String("32");`

B. `Exception e=new StackOverflowError("212");`

C. `Object o=new Math();`

D. `List h=new HashSet();`

E. None of the above are legal instantiations.

QUESTION 28

What could replace **`<1*>`** in the code to the the right so that method `funtime` works as intended?

A. `int[] a` B. `int a`

C. `int...a` D. A and C

E. None of the above will work.

QUESTION 29

What is demonstrated by the the section marked **`<1*>`**?

A. actual parameters

B. instance fields

C. overriding

D. overloading

E. formal parameters

QUESTION 30

Assuming **`<1*>`** is filled correclty, what is the output by the code to the right?

A. `4 20`

B. `5 15`

C. `5 20`

D. `4 15`

E. There is no output die to an error.

```
public static int funtime(<1*>)
{
  int tot=0;
  for(int h:a) {
    for(int k=1;k<=h;k*=2)
      tot++;
  }
  return tot;
}

//////////client code////////////////
////////////////////////////////////
out.print(funtime(16)+" ");
out.println(funtime(8,8,8,8,8));
```

```
class Node{
  int val, le,r;
  Node left,right;
  public Node(int v) {
          val =v;
  }
  public void print() {
          if(left<1*>)left.print();
          System.out.print(" "+val);
          if(right<1*>)right.print();
  }
  public int add(int v) {
          if(v>val) {
            if(right==null) {
                  right=new Node(v);
                  r++;
                  return 1+le;
            }
            else {
                  r++;
                  return le+1+right.add(v);
            }
          }
          if(v<=val) {
            if(left==null) {
                  left=new Node(v);
                  le++;
                  return 0;
            }
            else {
                  le++;
                  return left.add(v);
            }
          }
          return 0;
  }
}
/////client code//////////////////////////////////////////////////////////////////
int[]k= {3,5,2,5,7,4,6,5,9,12};
Node root=new Node(3);
for(int x=0;x<k.length;x++) {
 root.add(k[x]);
}
out.println(root.add(6)); //1
root.print(); //2
```

QUESTION 31

Which of the following could replace **<1*>** in the code above so that the `Node` class works as intended?

A. `==null`

B. `!=null`

C. `!=0`

D. All of the above.

E. B and C

QUESTION 32

Assuming that **`<*1>`** is filled correclty, what is the output by the line in the client code marked `//1`?

A. 4
B. 5
C. 6
D. 7
E. 8

QUESTION 33

What is the output by the line in the client code marked `//2`?

A. 2 3 3 4 5 5 5 6 6 7 9 12
B. 2 3 5 4 5 6 6 12 9 7 5 3
C. 3 3 2 5 5 4 5 7 6 6 9 12
D. 3 5 7 9 12 6 6 5 4 5 3 2
E. 12 9 7 6 6 5 5 5 4 3 3 2

QUESTION 34

Which data structure is implemented by the `Structure` class?

A. Red Black Tree
B. Max-Heap
C. Binary Search Tree
D. Priority Queue
E. Trie

QUESTION 35

What is output by the code to the right?

A. `truetrue`
B. `truefalse`
C. `falsetrue`
D. `falsefalse`
E. There are no errors on the code to the right.

```
Integer a=127;
Integer b=127;
out.print(a==b);
Integer c=420;
Integer d=420;
out.print(c==d);
```

QUESTION 36

Using the adjancey matrix on the right, which of the following is not an adjacent pair?

A. BA

B. AD

C. EA

D. CD

E. BC

QUESTION 37

How many edges exist in the graph depicted by the adjacency matrix to the right?

A. 5

B. 6

C. 7

D. 9

E. 8

	A	B	C	D	E
A	0	0	0	1	0
B	1	0	1	0	0
C	0	0	0	0	0
D	0	1	0	0	1
E	1	0	1	0	0

QUESTION 38

Draw the graph depicted by the above adjacency matrix.

QUESTION 39

What decimal value is represented by the following 8–bit two's complement number ?

10111011

QUESTION 40

What is the most efficient sort for worst case (exclude radix and counting sort)?

Name _____ Final Score _____ _____

School _____ 2nd Grading _____ _____

Division: 1A 2A 3A 4A 5A 6A Grade: 9 10 11 12 1st Grading _____ _____
score initials

The Virtual Challenge Meets
Computer Science - Student Answer Sheet

1)	_____	21)	_____
2)	_____	22)	_____
3)	_____	23)	_____
4)	_____	24)	_____
5)	_____	25)	_____
6)	_____	26)	_____
7)	_____	27)	_____
8)	_____	28)	_____
9)	_____	29)	_____
10)	_____	30)	_____
11)	_____	31)	_____
12)	_____	32)	_____
13)	_____	33)	_____
14)	_____	34)	_____
15)	_____	35)	_____
16)	_____	36)	_____
17)	_____	37)	_____
18)	_____	38)	_____
19)	_____	39)	_____
20)	_____	40)	_____

Total Number Correct _____ x 6 = _____

Total Number Incorrect _____ x -2 = _____

Final Score (record here and top right) _____

2017-2018 Virtual Challenge SQ Meet
Computer Science - KEY

1) C
2) A
3) C
4) B
5) B
6) E
7) A
8) D
9) B
10) C

11) B
12) C
13) C
14) A
15) C
16) C
17) A
18) A
19) A
20) A

21) D
22) C
23) B
24) A
25) D
26) C
27) E
28) C
29) E
30) C

31) B
32) D
33) A
34) C
35) B
36) D
37) C
38) see attached page
39) -69
40) mergesort/cubesort/timsort/heapsort

Note to Graders:

- All provided code segments are intended to be syntactically correct, unless otherwise stated (e.g. error is an answer). **Ignore any typographical errors.**

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- Any necessary Standard Java 2 Packages are assumed to have been imported as needed.
- Assume any undefined (undeclared) variables have been defined as used.

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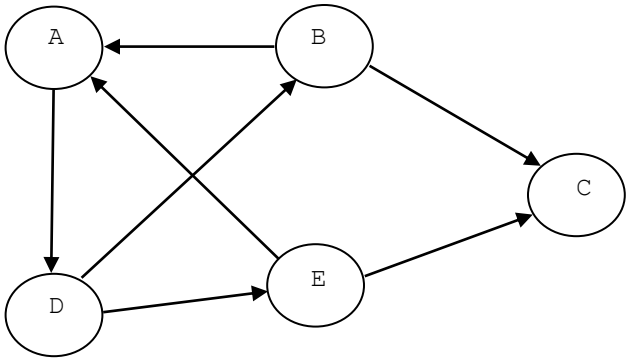
38)



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